

Sorting Methods

Time limit: 1.00 s

Memory limit: 512 MB

Here are some possible methods using which we can sort the elements of an array in increasing order:

1. At each step, choose two adjacent elements and swap them.
2. At each step, choose any two elements and swap them.
3. At each step, choose any element and move it to another position.
4. At each step, choose any element and move it to the front of the array.

Given a permutation of numbers $1, 2, \dots, n$, calculate the minimum number of steps to sort the array using the above methods.

Input

The first input line contains an integer n .

The second line contains n integers describing the permutation.

Output

Print four numbers: the minimum number of steps using each method.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$

Example

Input	Output
8 7 8 2 6 5 1 3 4	20 6 5 6